

Suvinava Basak

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Professional Experience

- Mar '25 – ■ **Student Hilfswissenschaftler** — *TU Braunschweig, Institute for Statistics and Foundations of Machine Learning, Germany*
- Conducting research on calibration and uncertainty estimation in modern neural networks under the supervision of Prof. Nicole Mücke.
- Aug '25 – ■ **Student Hilfswissenschaftler** — *Medizinische Hochschule Hannover, Germany*
- Designed and implemented a **multi-agent AI framework** powered by LLMs and Knowledge Graph-aware RAG for drug hypothesis generation.
- Apr '25 – Sep '25 ■ **Student Hilfswissenschaftler** — *TU Braunschweig, Institut für Analysis und Algebra, Germany*
- Graded student assignments and led tutorials for *Mathematical Algorithms and Programming 2* (SS 2025) for ~ 50 students.
- Aug '24 – Feb '25 ■ **Intern, Data Analytics & AI** — *Volkswagen Financial Services AG, Braunschweig, Germany*
- Developed an ML pipeline to automate invoice analysis and approval within vehicle maintenance workflows, reducing manual review time.
 - Designed and implemented a multi-stage Docker setup to optimize container security and reduce build time for deployment pipelines.
- Apr '18 – Oct '23 ■ **Strategy Consultant** — *Wipro Ltd., Bangalore, India*
- Conducted innovation opportunity research in AI/ML, driving concept-to-implementation.
 - Scouted strategic startup partners in Gen AI/ML and led alliances from evaluation to joint PoC development and Go-To-Market planning.
- Mar '15 – Jun '16 ■ **Solution Integrator** — *Ericsson Global India Services Pvt. Ltd., Kolkata, India*
- Configured and integrated a billing system for a leading US-based mobile carrier, ensuring end-to-end system consistency.
 - Analyzed historical usage patterns to design billing plans, improving alignment between client offerings and user behavior.

Education

- Oct '23 – ■ **M.Sc. Data Science**, *Technische Universität Braunschweig, Germany*
Planned thesis title: *End-to-End Calibration - Theory and Empirical Evaluation in Biomedical Context*.
- Jul '16 – Mar '18 ■ **Master of Business Administration**, *Indian Institute of Management Indore, India*
- Aug '10 – Jun '14 ■ **B.Tech. Computer Science & Engineering**, *West Bengal University of Technology, India*
Project title: *Face Recognition System using MATLAB*.

Projects

- **Covariate Shift Regularisation with Kernel** *(Gradient Descent, Linear Regression)*
Investigated the effect of covariate shift on model generalisation in linear regression using gradient descent; proposed and evaluated mitigation strategies based on importance weighting estimated through kernel density estimation and self-attention mechanisms.
- **Binary AMR Prediction from PBP4 using Transformers** *(DNABERT, Nucleotide Transformer, PyTorch)*
As part of the seminar "Deep Learning in Molecular Biology", benchmarked encoder-only pretrained transformer models for binary AMR prediction in *Staphylococcus aureus* against Cefoxitin resistance using *pbp4* gene sequences, evaluating classification performance across model architectures.
- **Multimodal Drug Synergy & Toxicity Prediction (Architecture Design)** *(GAT, ESM-2, Multi-omics)*
Designed a multimodal deep learning architecture integrating Graph Attention Networks on molecular SMILES representations, protein language model embeddings (ESM-2), gene expression omics data, and protein-protein interaction networks for joint drug synergy and toxicity prediction. Work conducted at Medizinische Hochschule Hannover; implementation discontinued following change in research group direction.

Publications

Journal / Preprint Articles

- 1 **S. Basak**, "Is grokking a loss of normal hyperbolicity of the interpolation manifold?" *arXiv preprint*, 2026, Accepted at SKILL 2026 (Studierendenkonferenz Informatik), Gesellschaft für Informatik e.V., Dresden.
- 2 R. Younis, **S. Basak**, L. Chavez, and Z. Ahmadi, "From literature to hypotheses: An AI co-scientist system for biomarker-guided drug combination hypothesis generation," *arXiv preprint*, 2026. [URL: https://arxiv.org/abs/2603.00612](https://arxiv.org/abs/2603.00612).

Technical Skills

Languages	■ Python • Julia • Java • SQL
Frameworks	■ PyTorch • TensorFlow • LangChain • Hugging Face • scikit-learn
AI / ML	■ Large Language Models (LLM) • Retrieval-Augmented Generation (RAG) • Multi-Agent Systems • Knowledge Graphs
DevOps	■ Docker • Git • Linux
Data & BI	■ Power BI • Tableau • Pandas • NumPy
Spoken	■ English (proficient) • Bengali (native) • Hindi (proficient) • German (beginner)

Awards & Recognition

- 2025/26 ■ **Deutschlandstipendium** — merit-based scholarship for high-achieving students.
- 2023 ■ **Customer Centricity Award**, Wipro Ltd. — Q1 FY2023.
- 2019 ■ **Excellence Award**, Wipro Ltd. — Q3–Q4 FY2019.
- 2016 ■ **Spot Award**, Ericsson Global India Services — Q3 FY2016.

References

Available on request.